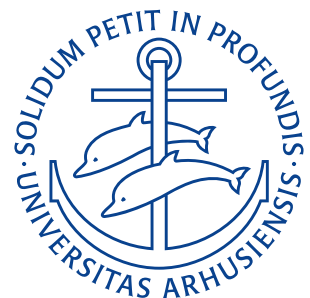


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Take-Home Exam

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Course: Numerical Analysis for Engineers — 290211U005
Date: May 17, 2026



Problem 1

Solve the following boundary value problem using the direct method.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = f(x, y) \quad (1)$$

where

$$f(x, y) = e^{-\frac{(x-15)^2}{4}} + e^{-\frac{(y-15)^2}{4}}. \quad (2)$$

Boundary conditions at the edges are as follows

$$\begin{aligned} u(0, y) &= 0 & u(L_x, y) &= 20 \\ \frac{\partial u}{\partial y} \Big|_{y=0} &= 0 & \frac{\partial u}{\partial y} \Big|_{y=L_y} &= 1. \end{aligned}$$

Using central difference scheme for all derivatives try to find the steady state response of the system. Discretize both x and y domain by 90 meshes for $L_x = L_y = 30$.

Note: Use the following to plot the calculated u in 3D:

```
surface(X,Y,u');  
view(3)
```

Derivation

We are told to use 90 meshes in both directions

$$L_x = L_y = 30 \quad (3)$$

which corresponds to a step-length of

$$h = \frac{30}{90} = \frac{1}{3} \quad (4)$$

and

$$N = 90 + 1 = 91 \quad (5)$$

grid points, corresponding to $x_i = ih, i = 0, 1, 2, \dots, 89, 90$ and $y_j = jh, j = 0, 1, 2, \dots, 89, 90$. As we are using the direct method, we will seek to solve the discrete approximation

$$Au = b \quad (6)$$

where

$$u = \begin{bmatrix} U_{1,1} \\ U_{2,1} \\ U_{3,1} \\ \vdots \end{bmatrix} \quad (7)$$

is a vector containing all unknown grid values of u , and A contains all the finite-difference stencil coefficients. We have in total $91^2 = 8281$ grid points, meaning the A -matrix must be of size 8281×8281 .

At any interior grid point, we can approximate the second derivative in x , y and the derivative in x and y as

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{h^2} \quad (8)$$

$$\frac{\partial^2 u}{\partial y^2} \approx \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{h^2} \quad (9)$$

$$\frac{\partial^2 u}{\partial x \partial y} \approx \frac{U_{i+1,j+1} - U_{i+1,j-1} - U_{i-1,j+1} + U_{i-1,j-1}}{4h^2}. \quad (10)$$

Inserting eqs. (8) to (10) into eq. (1), and defining $f_{i,j} = f(x_i, y_i)$ yields:

$$\begin{aligned} \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{h^2} + \frac{U_{i+1,j+1} - U_{i+1,j-1} - U_{i-1,j+1} + U_{i-1,j-1}}{4h^2} + \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{h^2} &= f_{i,j} \\ U_{i+1,j} - 2U_{i,j} + U_{i-1,j} + \frac{1}{4}U_{i+1,j+1} - \frac{1}{4}U_{i+1,j-1} - \frac{1}{4}U_{i-1,j+1} + \frac{1}{4}U_{i-1,j-1} + U_{i,j+1} - 2U_{i,j} + U_{i,j-1} &= h^2 f_{i,j} \\ 4U_{i,j} - U_{i+1,j} - U_{i-1,j} - U_{i,j+1} - U_{i,j-1} - \frac{1}{4}U_{i+1,j+1} + \frac{1}{4}U_{i+1,j-1} + \frac{1}{4}U_{i-1,j+1} - \frac{1}{4}U_{i-1,j-1} &= -h^2 f_{i,j}. \end{aligned} \quad (11)$$

As a stencil, this looks like

$$\begin{array}{ccc} \frac{1}{4} & -1 & -\frac{1}{4} \\ -1 & 4 & -1 \\ -\frac{1}{4} & -1 & \frac{1}{4} \end{array}.$$

The left boundary condition is a Dirichlet-type

$$u(0, y) = 0 \implies U_{0,j} = 0, j = 0, 1, 2, \dots, 89, 90 \quad (12)$$

and the right boundary condition is also a Dirichlet-type boundary condition

$$u(L_x, y) = 20 \implies U_{90,j} = 20, j = 0, 1, 2, \dots, 89, 90. \quad (13)$$

At the bottom and top we are given Von Neumann boundary conditions. At the bottom,

$$\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0$$

now, using a central difference at $j = 0$ we get

$$\frac{U_{i,1} - U_{i,-1}}{2h} = 0. \quad (14)$$

In eq. (14) the point $U_{i,-1}$ is outside of the region, we however quickly see that for eq. (14) to hold, $U_{i,1} = U_{i,-1}$. That means the bottom boundary condition becomes

$$\begin{aligned} \frac{U_{i+1,0} - 2U_{i,0} + U_{i-1,0}}{h^2} + \frac{2U_{i,1} - 2U_{i,0}}{h^2} &= f_{i,0} \\ 4U_{i,0} - U_{i+1,0} - U_{i-1,0} - 2U_{i,1} &= h^2 f_{i,0} \quad i = 1, \dots, 89. \end{aligned} \quad (15)$$

Following a similar procedure at the top we get:

$$\begin{aligned} \left. \frac{\partial u}{\partial y} \right|_{y=L_x} &= 1 \\ \frac{U_{i,91} - U_{i,89}}{2h} &= 1 \\ U_{i,91} &= U_{i,89} + 2h \\ \frac{U_{i+1,90} - 2U_{i,90} + U_{i-1,90}}{h^2} + \frac{2U_{i,89} - 2U_{i,90} + 2h}{h^2} &= f_{i,90} \\ 4U_{i,90} - U_{i+1,90} - U_{i-1,90} - 2U_{i,89} &= -h^2 f_{i,90} + 2h \\ &= -h^2 f_{i,90} + \frac{2}{3}. \end{aligned} \quad (16)$$

Based on this, the Matlab code attached in NumAna_NoahRBH_2026_prob_1.m has been written. Running this code results in fig. 1.

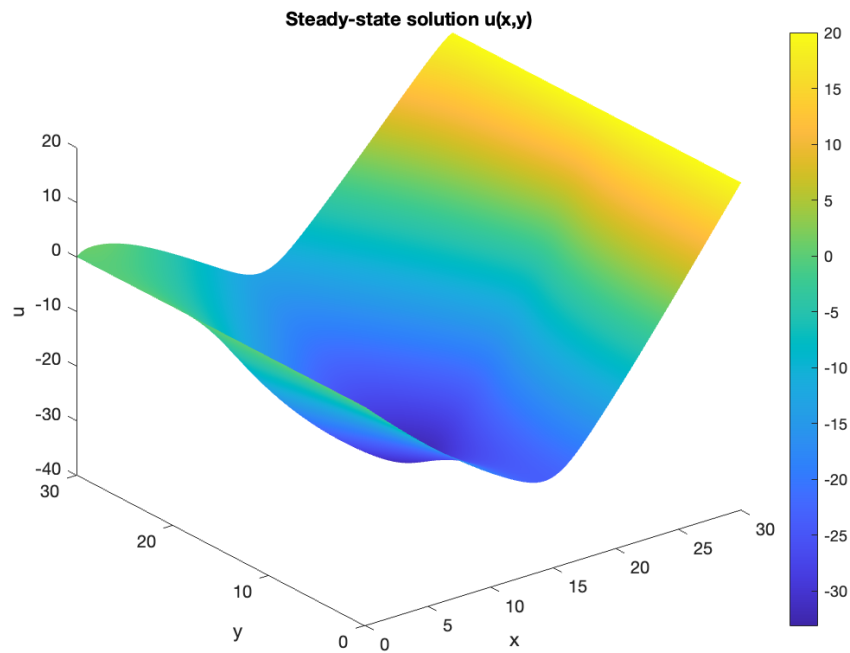


Figure 1: Steady-state solution to Problem 1, from Matlab using the direct method.

Problem 2

In medical physics, the transport of a drug through a specific segment of an artery is modeled using the Advection-Diffusion-Reaction (ADR) Equation:

$$\frac{\partial u}{\partial t} + \alpha \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2} - cu.$$

In this model

- α represents the speed of blood flow. It is the primary mechanism that physically “pushes” the drug particles downstream through the arterial segment.
- ν is the Drug Dispersion Coefficient which accounts for the spreading of drug concentration. It combines molecular diffusion with “Taylor dispersion”, which is the mixing effect caused by the blood’s velocity profile as it interacts with the vessel walls.
- c is the Drug Concentration Decay Rate representing a linear sink term. In pharmacokinetics, this models how the drug is “lost” from the system, typically through metabolic breakdown by the liver or absorption into the surrounding vascular tissues.

Assume the blood stream velocity $\alpha = 1 \frac{\text{cm}}{\text{s}}$, drug dispersion coefficient $\nu = 0,1 \text{ cm}^2/\text{s}$, and drug concentration decay rate $c = 0,05 \text{ s}^{-1}$. We want to study the ADR for 20 seconds within 20 cm length of the artery from where the drug is injected considering the following initial and boundary conditions:

- **Initial Conditions:** at $t = 0$ an initial distribution of drug exists in the artery

$$u(x, 0) = e^{-(x-5)^2}.$$

- **Boundary Conditions:** at $x = 0$ a medical pump provides a variable dosage

$$u(0, t) = \sin^2(t)$$

at $x = 20$ a “free-exit” condition where the drug leaves the segment without resistance

$$\left. \frac{\partial u}{\partial x} \right|_{x=20} = -0,2.$$

Using implicit Crank-Nicholson Scheme try to solve the problem. Use a time step of $\Delta t = 0,01$ to discretize the time domain and 100 mesh to discretize the space domain (number of grid points in space: $N = 101$).

Animate the results for 20 seconds and plot $u(x, y)$ at the time $t = 2, t = 5, t = 7, t = 10, t = 12, t = 15, t = 18$, and $t = 20$ s.

Derivation

The segment of the artery which we are modelling is

$$0 \leq x \leq 20 \quad (17)$$

for times

$$0 \leq t \leq 20. \quad (18)$$

A mesh of 100 should be employed corresponding to

$$N = 100 + 1 = 101 \quad (19)$$

grid points. The spatial step-size is therefore:

$$h = \Delta x = \frac{20 - 0}{101 - 1} = 0,2 \quad (20)$$

and the time step is

$$\Delta t = 0,01, \quad (21)$$

which corresponds to

$$N_t = \frac{20}{0,01} = 2000 \quad (22)$$

time steps. Based on eqs. (19) and (20), we define the grid points

$$x_i = (i - 1)h, \quad i = 1, 2, \dots, N \quad (23)$$

and based on eqs. (21) and (22), we define the time points

$$t_n = n\Delta t, \quad n = 0, 1, \dots, N_t. \quad (24)$$

We will seek to approximate the solution of $u(x, t)$ by a discrete table of values

$$u_i^n = u(x_i, t_n). \quad (25)$$

At t_0 our initial condition is,

$$u(x, 0) = e^{-(x-5)^2} \quad (26)$$

which on the form of eq. (25) is

$$u_i^0 = e^{-(x-5)^2}.$$

The left boundary condition is a Dirichlet boundary condition prescribing

$$u(0, t) = \sin^2(t) \quad (27)$$

on the same form as eq. (25) this corresponds to

$$u_1^n = \sin^2(t).$$

At $x = 20$, we have a Von Neumann boundary condition,

$$\left. \frac{\partial u}{\partial x} \right|_{x=20} = -0,2. \quad (28)$$

Since $x = 20$ we have to use something other than the central difference if we want to avoid points outside our defined interval (as was the case for Problem 1). Using a Taylor-expansion the second-order backward difference approximation is

$$u_x(x_N) = \frac{3u_N^n - 4u_{N-1}^n + u_{N-2}^n}{2h}.$$

So our Von Neumann boundary condition becomes

$$\frac{3u_N^n - 4u_{N-1}^n + u_{N-2}^n}{2h} = -0,2 \implies 3u_N^n - 4u_{N-1}^n + u_{N-2}^n = -0,08.$$

For the interior points, i.e. $i = 2, 3, \dots, N-1$, we approximate the first derivative by a central difference

$$u_x(x_i, t_n) \approx \frac{u_{i+1}^n - u_{i-1}^n}{2h} \quad (29)$$

and the second derivative, we approximate by

$$u_{xx}(x_i, t_n) \approx \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{h^2}. \quad (30)$$

Now, rewriting the given PDE as

$$\frac{\partial u}{\partial t} = -\alpha \frac{\partial u}{\partial x} + v \frac{\partial^2 u}{\partial x^2} - cu \quad (31)$$

and substituting eqs. (29) and (30) into eq. (31) we get the spatial operator

$$L_h u_i = -\alpha \frac{u_{i+1} - u_{i-1}}{2h} + v \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} - cu.$$

So the PDE becomes:

$$\frac{du_i}{dt} = L_h u_i.$$

Now, we use Crank-Nicholson in time. Here, we average the spatial operator between the old and new time level,

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{1}{2} (L_h u_i^{n+1} + L_h u_i^n). \quad (32)$$

Rearranging eq. (32), we get

$$u_i^{n+1} - \frac{\Delta t}{2} L_h u_i^{n+1} = u_i^n + \frac{\Delta t}{2} L_h u_i^n.$$

For interior points, we define

$$\mu = \frac{v\Delta t}{2h^2} \quad (33)$$

$$\lambda = \frac{\alpha\Delta t}{4h} \quad (34)$$

$$\rho = \frac{c\Delta t}{2} \quad (35)$$

using the values given in the problem we obtain

$$\begin{aligned}\mu &= \frac{0,1 \cdot 0,01}{2 \cdot (0,2)^2} = 0,0125 \\ \lambda &= \frac{1 \cdot 0,01}{4 \cdot 0,2} = 0,0125 \\ \rho &= \frac{0,05 \cdot 0,01}{2} = 0,00025.\end{aligned}$$

The Crank-Nicholson equation for an interior point is

$$-(\mu + \lambda) u_{i-1}^{n+1} + (1 + 2\mu + \rho) u_i^{n+1} + (\lambda - \mu) u_{i+1}^{n+1} = (\mu + \lambda) u_{i-1}^n + (1 - 2\mu - \rho) u_i^n + (\mu - \lambda) u_{i+1}^n. \quad (36)$$

As $\mu = \lambda$, $\mu - \lambda = 0$, so the above simplifies to:

$$-(\mu + \lambda) u_{i-1}^{n+1} + (1 + 2\mu + \rho) u_i^{n+1} = (\mu + \lambda) u_{i-1}^n + (1 - 2\mu - \rho) u_i^n. \quad (37)$$